Clinic Appointment & Prescription System

GROUP 3 | Meghna, Santhoshi, Divya | Team size: 3 | Duration: 8 days

Backend	Frontend	Database	Auth	Deployment
Flask (REST API)	React + Vite	PostgreSQL + SQLAlchemy	JWT (access + refresh)	Docker Compose

1. Problem Statement

Patients search doctors by specialization and book a free slot; the doctor runs the consultation and issues a prescription that becomes part of the patient visit history.

2. Roles & Permissions

Role	What they can do
Admin	Manages doctors, specializations and the clinic calendar.
Doctor	Sets weekly availability, sees the day queue, records consultations and prescriptions.
Patient	Searches doctors, books/reschedules/cancels, views their own history.

3. Data Model

- **User** - credentials, role, name, phone.
- **DoctorProfile** - user, specialization, qualification, consultation fee, room.
- **AvailabilitySlot** - doctor, weekday, start time, end time, slot duration.
- **Appointment** - doctor, patient, date, start time, status (booked / completed / cancelled / no-show), reason.
- **Consultation** - appointment, vitals, complaint, diagnosis notes, follow-up date.
- **Prescription** + **PrescriptionItem** - consultation, medicine, dosage, frequency, duration, instructions.
- **Medicine** - name, form, strength (master list used for autocomplete).

4. Functional Requirements

- Doctor availability defined by weekday and time range; the API generates concrete bookable slots for a requested date.
- Patient search by specialization, doctor name, fee range and available date.
- Book, reschedule and cancel; status transitions enforced on the server.
- Doctor's day queue sorted by slot time, with a one-click open consultation.
- Consultation form: vitals, complaint, diagnosis notes, follow-up date.
- Multi-medicine prescription with dosage and duration, printable.
- Patient visit history timeline with past prescriptions.
- Admin: add doctors, manage specializations, view clinic-wide appointment counts.

5. Core Technical Challenge

No double booking and strict record isolation. Slot generation subtracts already-booked appointments, and the booking endpoint re-checks inside a transaction with a unique constraint on (doctor, date, start_time) so two simultaneous requests cannot take the same slot. Cancellation within 2 hours of the slot is rejected. A patient can read only their own appointments, consultations and prescriptions; a doctor only those of patients they have treated - enforced in the query, not by hiding buttons in React.

6. React Screens

- Doctor search with filters and result cards.
- Calendar date picker with available slots for the chosen doctor.
- My appointments (upcoming / past) and patient history timeline.
- Doctor day queue and consultation screen with prescription builder.
- Admin doctor management panel.

7. Authentication & Authorisation Specification

- Passwords hashed with bcrypt - plain text passwords must never touch the database.
- Endpoints: `POST /auth/register`, `/auth/login`, `/auth/refresh`, `/auth/logout`, `GET /auth/me`.
- The access token carries a `role` claim; a custom decorator `@role_required("admin")` guards privileged routes on top of `@jwt_required()`.
- Refresh rotation: a refresh call issues a new access token; logout adds the token JTI to a blocklist so it cannot be reused.
- React side: Axios request interceptor attaches the token, response interceptor catches 401, calls refresh once, retries the original request, and logs the user out if refresh also fails.
- Route protection in React by role - a seeker must never be able to open an admin screen, even by typing the URL.

8. Docker Specification

- `backend/Dockerfile` - Python slim base, requirements installed first for layer caching, served by `gunicorn` (not the Flask dev server).
- `frontend/Dockerfile` - multi-stage: Node stage runs `npm run build`, nginx stage serves the static build and proxies `/api` to the backend.
- `docker-compose.yml` - services `api`, `web`, `db` on one network, a named volume for database data, a healthcheck on `db`, and `depends_on` so the API waits for it.
- All secrets (DB URL, JWT secret key) come from `.env`; `.env.example` is committed, `.env` is not.
- Acceptance test: on a clean machine, `git clone` then `docker compose up --build` gives a working app with seeded demo data.

9. Phase-wise Execution Plan

Phase 1 - Day 1: Foundation

- Finalise the data model and draw the ER diagram / collection diagram before writing code.
- Repo and folder structure: `backend/`, `frontend/`, `docker-compose.yml`, `README.md`.
- Docker skeleton up early - database container running and reachable from the API container on day one.
- User model + full JWT auth flow + role seeding. Do not move on until login and refresh work in Postman.

Phase 2 - Day 2 to 3: Core API

- Every core CRUD endpoint with request validation and clear error responses (400 / 401 / 403 / 404 / 409).
- Role guards on every write route; ownership checks so a user can only touch their own records.
- Pagination, filtering and sorting on all list endpoints - never return the whole table.
- Test every endpoint in Postman and save the collection before you write a single React component.

Phase 3 - Day 3 to 5: React Application

- Routing, auth context, protected route wrapper, login and register screens.
- List, detail and create/edit screens for the main entities with client-side form validation.
- Loading, empty and error states on every screen - no blank page while data is fetching.
- Shared Axios instance; no API URL hardcoded in components.

Phase 4 - Day 5 to 6: Core Business Logic

- Implement the core rule described above - this is what separates this project from a CRUD demo.
- Dashboard and aggregation endpoints computed in the database layer, not in JavaScript.
- Search and filters wired end to end.

Phase 5 - Day 6 to 8: Harden and Ship

- Full dockerization verified from a clean clone; fix anything that only worked on your laptop.
- Seed script that creates demo users for each role plus realistic sample data.
- README: architecture overview, setup steps, environment variables, API endpoint table, demo credentials.
- Postman collection, bug fixing pass, and a 5-minute demo recording walking through each role.

10. Work Split

Owner	Responsibility
Member 1	Auth, roles, doctor profiles, availability and slot generation.
Member 2	Appointments, booking rules, consultations, prescriptions, medicine master.
Member 3	Entire React application for all three roles.

Every member must commit backend code and review at least one pull request - no single member owns the whole repository.

11. Deliverables

- GitHub repository containing both applications with a readable commit history from every member.
- README.md with setup instructions, environment variables and an API endpoint table.
- ER diagram (SQL) or collection diagram (MongoDB).
- Seed script with demo data and demo credentials per role.
- Postman collection covering every endpoint.
- Working `docker compose up --build` on a clean machine.
- 5-minute demo video.

12. Evaluation

Area	Weight	What is checked
Functionality	30%	All listed features work end to end without manual database edits

Area	Weight	What is checked
API design & JWT	20%	Correct status codes, validation, role guards, refresh and blocklist working
Core business logic	20%	The project-specific core rule is correctly and safely implemented
React frontend	15%	Component structure, state handling, validation, loading and error states
Docker & docs	15%	One-command startup, clean compose file, complete README and seed data

13. Stretch Goals (only after everything above works)

- Printable prescription PDF.
- Reminder table listing appointments due in the next 24 hours.